

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO2-AT1 - Scenario Based Programs

Programming in Java

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Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO2-AT1 - Scenario Based Programs

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JAVA PROGRAMMING

Q1 Student Registration System
10 marks

Q2 E Commerce Cart System
10 marks

Q3 Library Management System
10 marks

Q4 Employee Data Processing
10 marks

Q5 Online Voting System
10 marks

5/5 answered

Q1 Student Registration System
10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.
(a) Enumerate three Collection interfaces suitable for this system.
(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
(c) Write a Java program using HashSet and HashMap with Generics to:
• Add students
• Map students to courses
• Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 public class StudentRegistrationSystem {
4     public static void main(String[] args) {
5
6         HashSet<String> studentSet = new HashSet<>();
7
8         HashMap<String, List<String>> studentCourseMap = new HashMap<>();
9
10        studentSet.add("S101");
11        studentSet.add("S102");
12        studentSet.add("S103");
13        studentSet.add("S101");
14    }
15 }
```

Input
stdin input

Output

```
Unique Student IDs:
S102
S103
S101
Student - Course Relationship
```

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JAVA PROGRAMMING

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QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3
4 public class ECommerceCartSystem {
5     public static void main(String[] args) {
6
7         ArrayList<String> cart = new ArrayList<>();
8
9         cart.add("Laptop");
10        cart.add("Mobile Phone");
11        cart.add("Headphones");
12        cart.add("Keyboard");
13
14        System.out.println("Items added to cart successfully!");
```

Input

stdin input

Output

```
Items added to cart successfully!
Cart items before removal:
- Laptop
- Mobile Phone
- Headphones
```

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JAVA PROGRAMMING

QUESTIONS

Q1
Student Registration System
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10 marks

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Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:
• Insert book details
• Retrieve a book
• Display all books

Your Code

```
1 import java.util.HashMap;
2 import java.util.Map;
3
4 public class LibraryManagementSystem {
5     public static void main(String[] args) {
6
7         Map<String, BookDetails> library = new HashMap<>();
8
9         library.put("978-0134685991", new BookDetails("Effective Java",
10 library.put("978-0596009205", new BookDetails("Head First Design
11 library.put("978-0132350884", new BookDetails("Clean Code", "R
12
13         System.out.println("Books inserted successfully!\n");
14     }
15 }
```

Input

stdin input

Output

```
Books inserted successfully!
Book found for ISBN 978-0596009205:
Title: Head First Design
Patterns Author: Eric Freeman
```

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QUESTIONS

Q1

Student Registration System
10 marks

10 marks

Q2

F-Commerce Cart System

10 marks

Q3

Library Management System

8.0. on notes.

Q4

Employee Data Processing

Employees

05

Online Voting System

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5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- List three advantages of Generics in Collections.
- Represent how employee data is stored using List with Generics.
- Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2
3 public class EmployeeDataProcessing {
4     public static void main(String[] args) {
5
6         List<Employee> employeeList = new ArrayList<Employee>();
7
8         employeeList.add(new Employee(1, "Arun", 45000));
9         employeeList.add(new Employee(2, "Bala", 62000));
10        employeeList.add(new Employee(3, "Charan", 58000));
11        employeeList.add(new Employee(4, "Divya", 40000));
12        employeeList.add(new Employee(5, "Esha", 75000));
13
14        System.out.println("Employees with salary > 50,000:");
15    }
16 }
```

Input

```
stdin input
```

Output

```

Employees with salary > 50,000:
ID: 2, Name: Bala, Salary:
62000.0
ID: 3, Name: Charan, Salary:
58000.0
ID: 5, Name: Esha, Salary:

```

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Q5
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;
2
3 public class OnlineVotingSystem {
4     public static void main(String[] args) {
5
6         Set<String> votedUsers = new HashSet<>();
7
8         Map<String, Integer> voteCounts = new HashMap<>();
9
10        voteCounts.put("Alice", 0);
11        voteCounts.put("Bob", 0);
12        voteCounts.put("Charlie", 0);
13
14        String[][] incomingVotes = {
15            {"user1", "Alice"},
16            {"user2", "Bob"},
17            {"user3", "Charlie"},
18            {"user4", "Alice"},
19            {"user5", "Bob"},
20            {"user6", "Charlie"},
21            {"user7", "Alice"},
22            {"user8", "Bob"},
23            {"user9", "Charlie"},
24            {"user10", "Alice"}
25        };
26    }
27 }
```

Input

stdin input

Output

--- Processing Votes ---
Vote recorded successfully for Alice by user1
Vote recorded successfully for Bob by user2
Vote recorded successfully for Charlie by user3
Vote recorded successfully for Alice by user4
Vote recorded successfully for Bob by user5
Vote recorded successfully for Charlie by user6
Vote recorded successfully for Alice by user7
Vote recorded successfully for Bob by user8
Vote recorded successfully for Charlie by user9
Vote recorded successfully for Alice by user10

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